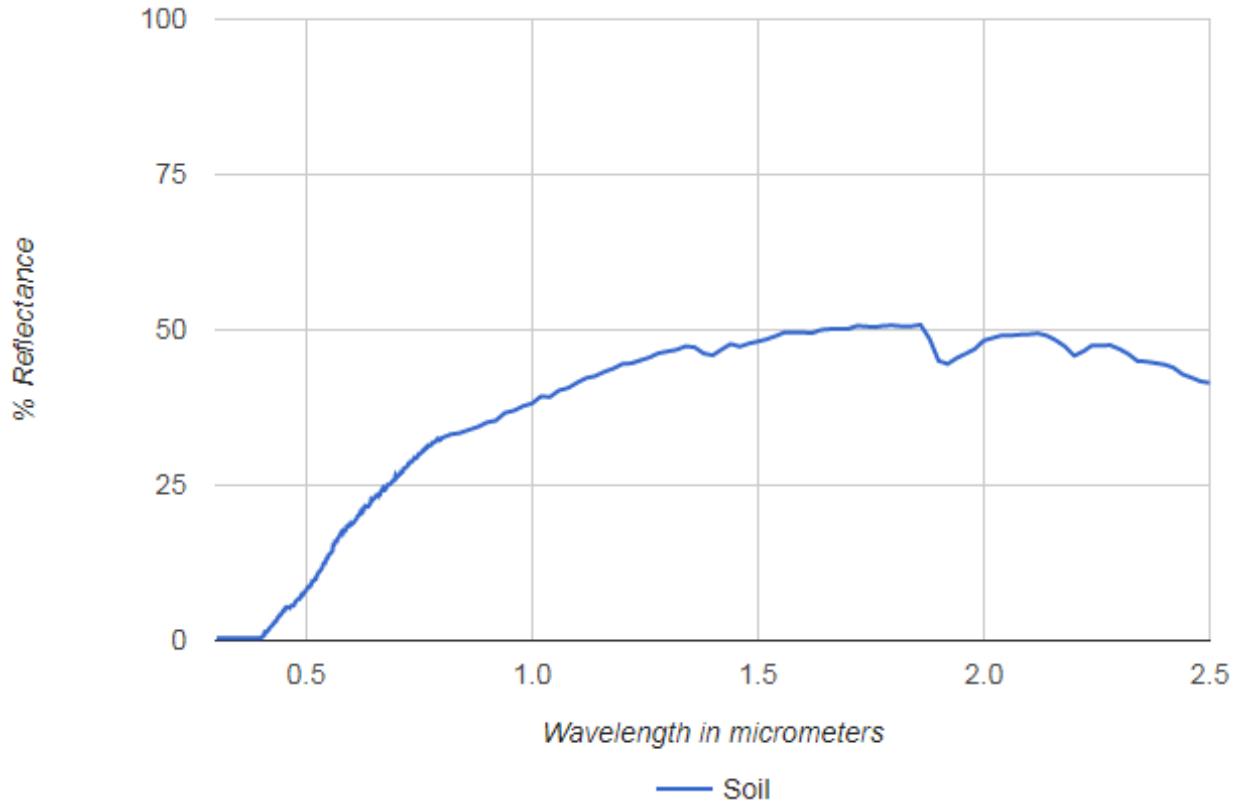

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

General Spectral Reflectance of Soil



Grain Size
Small
to
Large

Reflectance

Dry Soil Condition

Grain Size
Small
to
Large

Reflectance

Wet Soil Condition

SPECTRAL REFLECTANCE OF VEGETATION

INTERACTION OF EMR WITH VEGETATION

- Vegetation shows variations that depend upon the climate and soil
- Plants use solar energy to convert water and carbon dioxide to carbohydrates and oxygen through the process of photosynthesis, with a strong influence on their interaction with EMR

INTERACTION OF EMR WITH VEGETATION

- The catalyst for photosynthesis is the pigment chlorophyll
- The function of chlorophyll is to absorb solar radiations and thereby fuel photosynthesis
- This is achieved by absorption near **0.45 and 0.68 µm** in the blue and red part of the visible spectrum and reflectance in green, *which is why healthy leaves appear* ***green***

INTERACTION OF EMR WITH VEGETATION

- Chlorophyll, water content, and surface morphology of different plants have a marked effect on their spectral properties in the visible to NIR spectrum

INTERACTION OF EMR WITH VEGETATION

- Healthy vegetation looks very similar (green color) in the visible part of the spectrum but can be distinguished in NIR
- Plants' leaves, spaces, and branches with different leaf shapes and sizes can broaden the range of response, ease discrimination between species and healthy and weak members of the same species

INTERACTION OF EMR WITH VEGETATION

- Vegetation has life cycles of various durations; thus, its spectral properties are not fixed
- As a deciduous leaf matures before falling, its chlorophyll contents decay away, removing the strong red absorption; increasing reflectance and consequently, it changes its color from green to yellow to red
- As cell shrinks and dry, they become less efficient at reflecting NIR

INTERACTION OF EMR WITH VEGETATION

- As the leaves fall, less of the plant intercepts solar radiation, and the reflectance becomes dominated by the soil and leaf litter beneath it
- When new leaves appear, NIR reflectance is well developed

INTERACTION OF EMR WITH VEGETATION

- Coniferous trees do not shed their leaves during one season and always display high chlorophyll abundance
- Since their leaves are small, less solar radiation is intercepted, so the reflectance of the whole plant is lower than the deciduous trees with full leaf

Trees with leaves off in FALL



Trees with leaves on in FALL

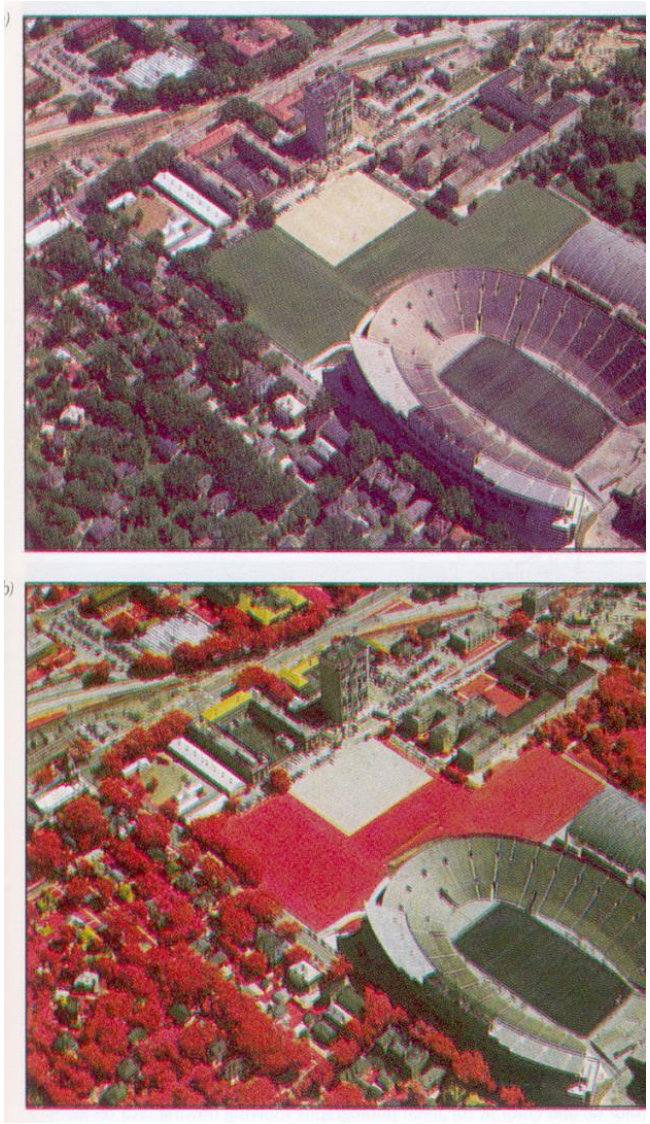


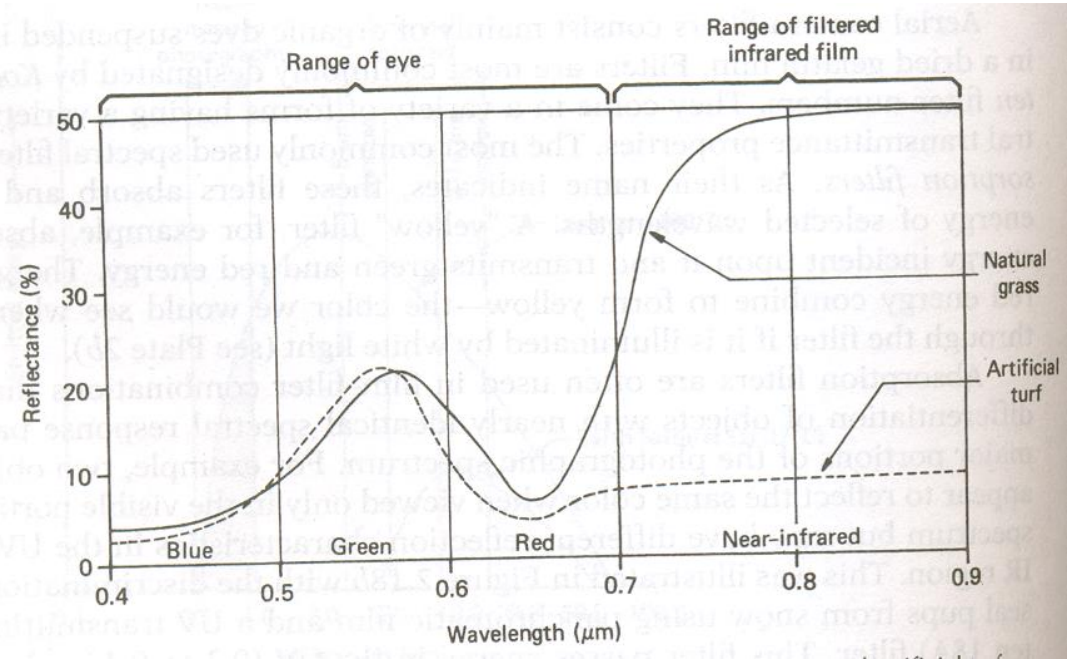
Spectral Variations



GREEN VEGITATION

- Vegetation reflectance is much higher in reflective IR region than visible
- This high IR reflectance is something unique to green vegetation (live) compared to artificial green (a camouflage detection property) or greenhouse





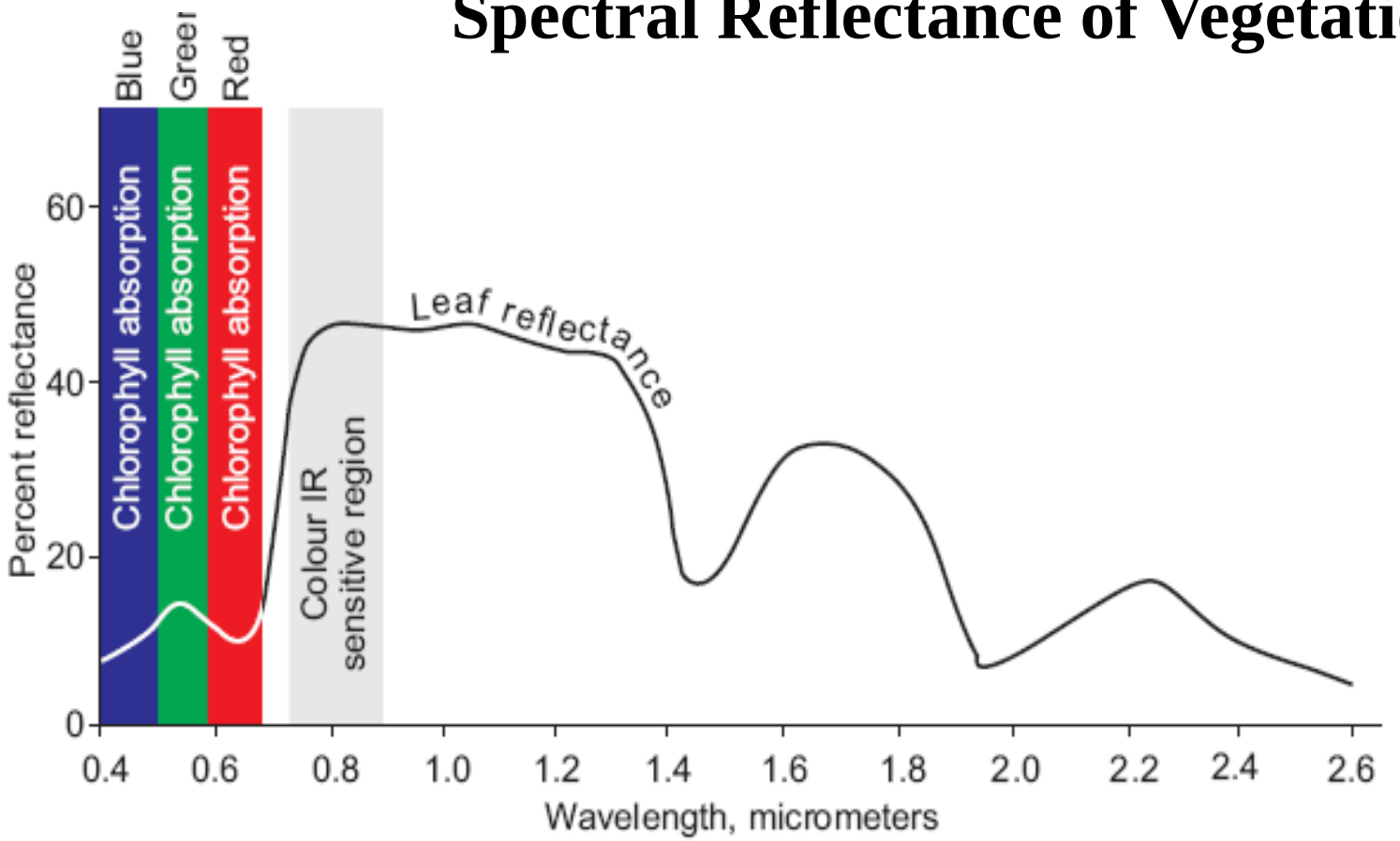
Spectral reflectance curves for natural grass and turf artificially made of green color to visually resemble green grass, reflectance in blue, red, and green is similar for both Surfaces

In NIR natural grass reflects very high, but turf does not

SPECTRAL REFLECTANCE CHARACTERISTICS

- Dead vegetation reflects a greater amount of energy than healthy green vegetation throughout the visible spectrum (0.4-0.7 µm)
- Conversely, it reflects less in the reflective NIR region, 0.7-1.1 µm

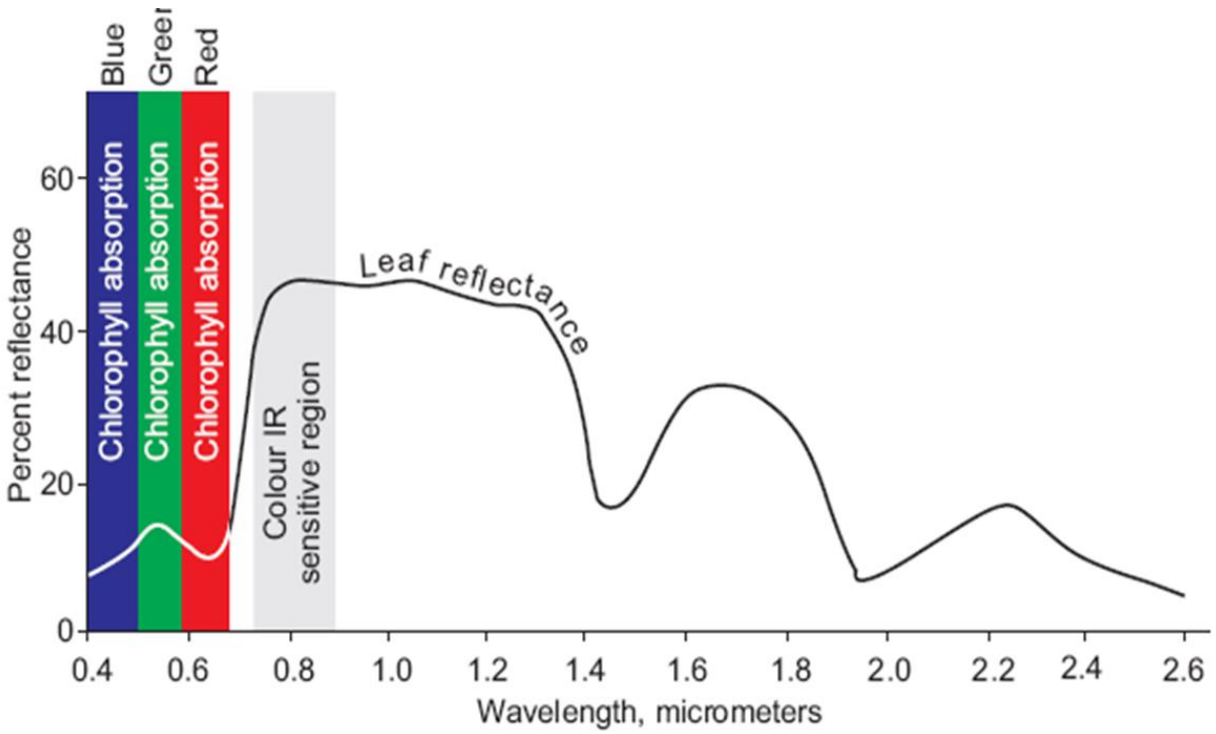
Spectral Reflectance of Vegetation



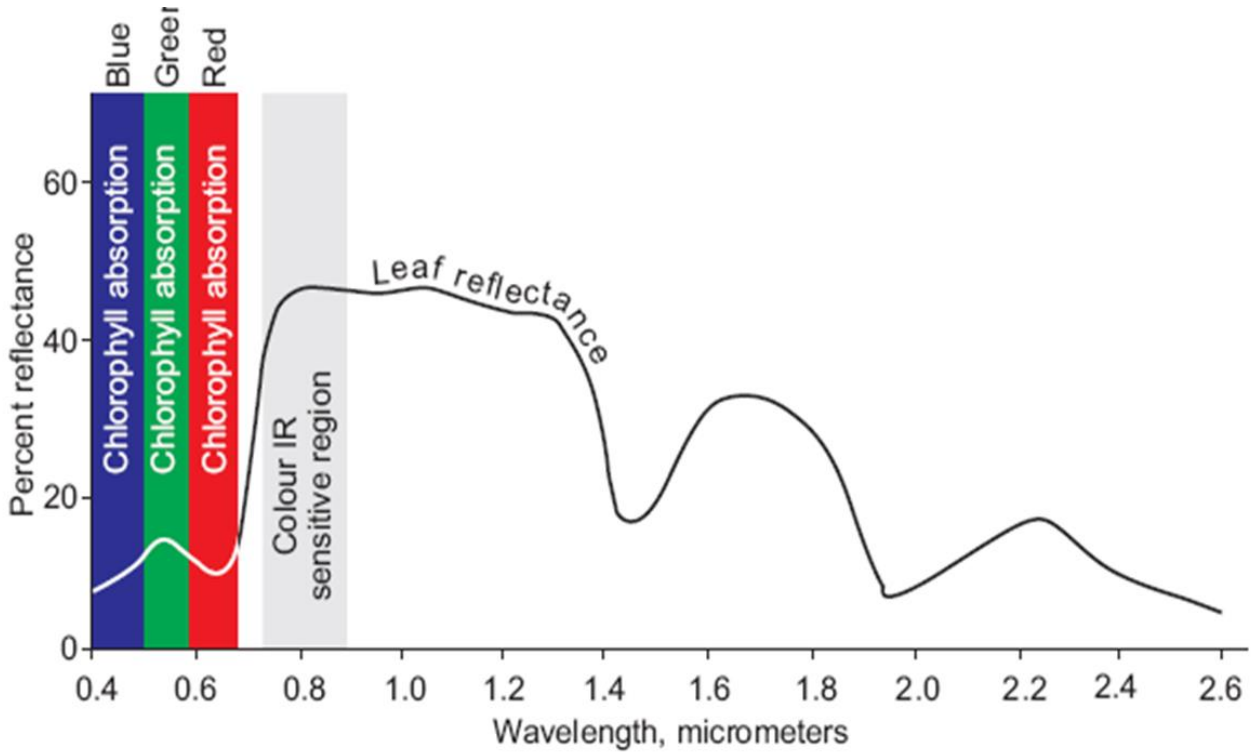
Strong absorption in visible and high reflectance in NIR

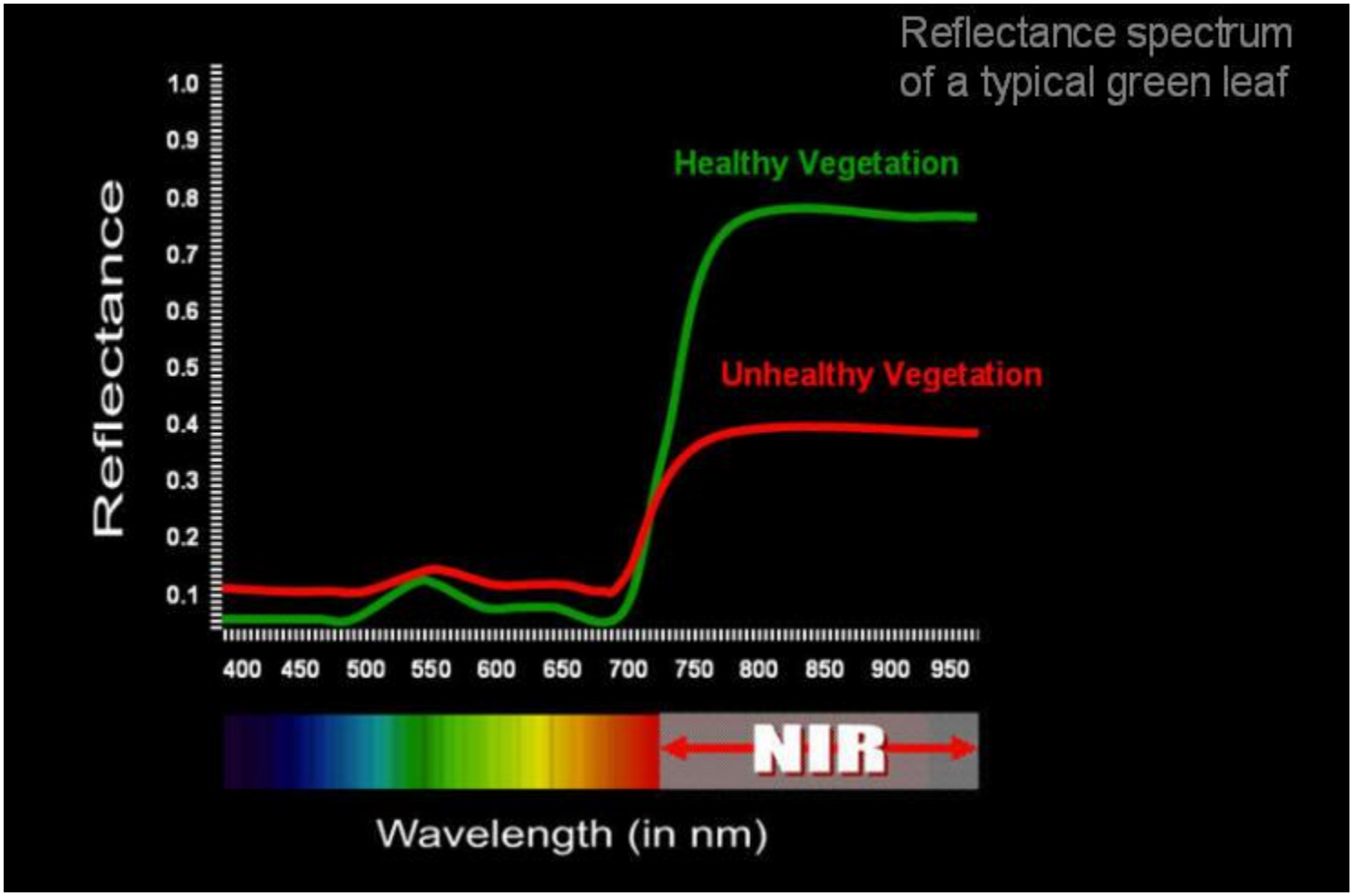
Chlorophyll strongly absorbs energy in the wavelength bands at about 0.45 and 0.67 μm , often called the "Chlorophyll Absorption Bands"

- From the visible to the NIR portion of the spectrum at about 0.7 μm , the reflectance of healthy vegetation increases dramatically
- In the range from about 0.7 to 1.3 μm , a plant leaf typically reflects 40 to 50 percent of the energy incident upon it

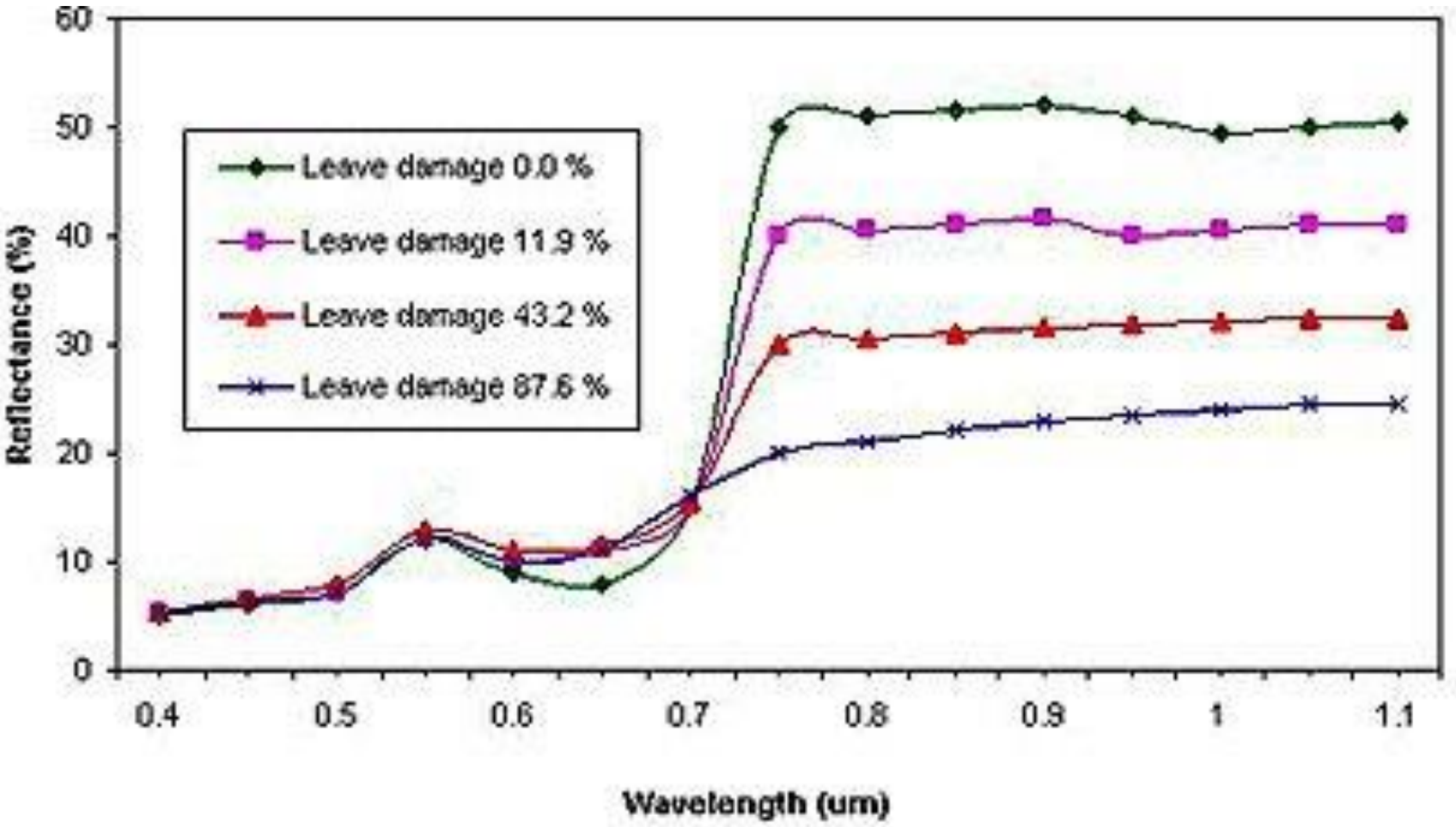


- If a plant is subject to some form of stress that interrupts its normal growth and productivity, it may decrease or cease chlorophyll production
- This result is less chlorophyll absorption in the blue and red bands. Often, the red reflectance increases to the point that we see the plant turn yellow (a combination of green and red)




Reflectance spectrum
of a typical green leaf
Reflectance
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
Healthy Vegetation
Unhealthy Vegetation
400 450 500 550 600 650 700 750 800 850 900 950
NIR
Wavelength (in nm)

Quantitatively in this set of field spectral measurements of leaves taken from soybean plants, increasing stress that causes loss of water and breakdown of cell walls



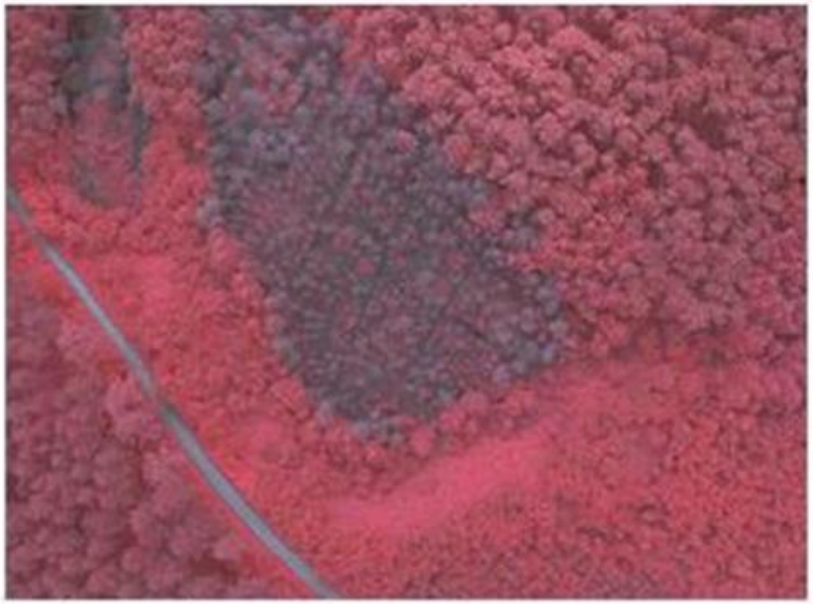
Spectral Reflectance - Vegetation

Remote sensed imagery can be used to detect stressed or diseased plants

- High NIR reflectance / Low visible reflectance = Healthy
- Low NIR reflectance / High visible reflectance = Unhealthy

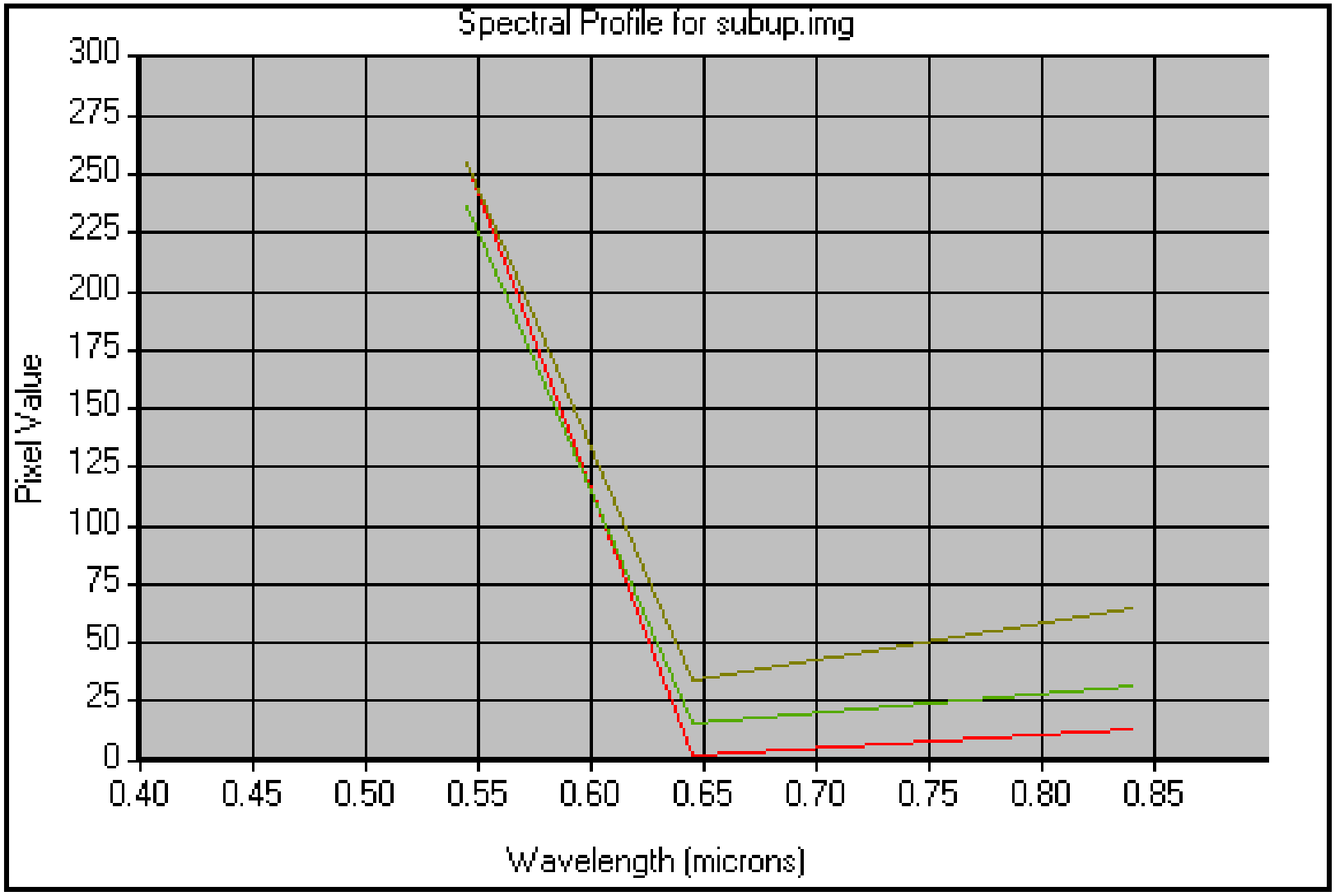


True Colour RGB



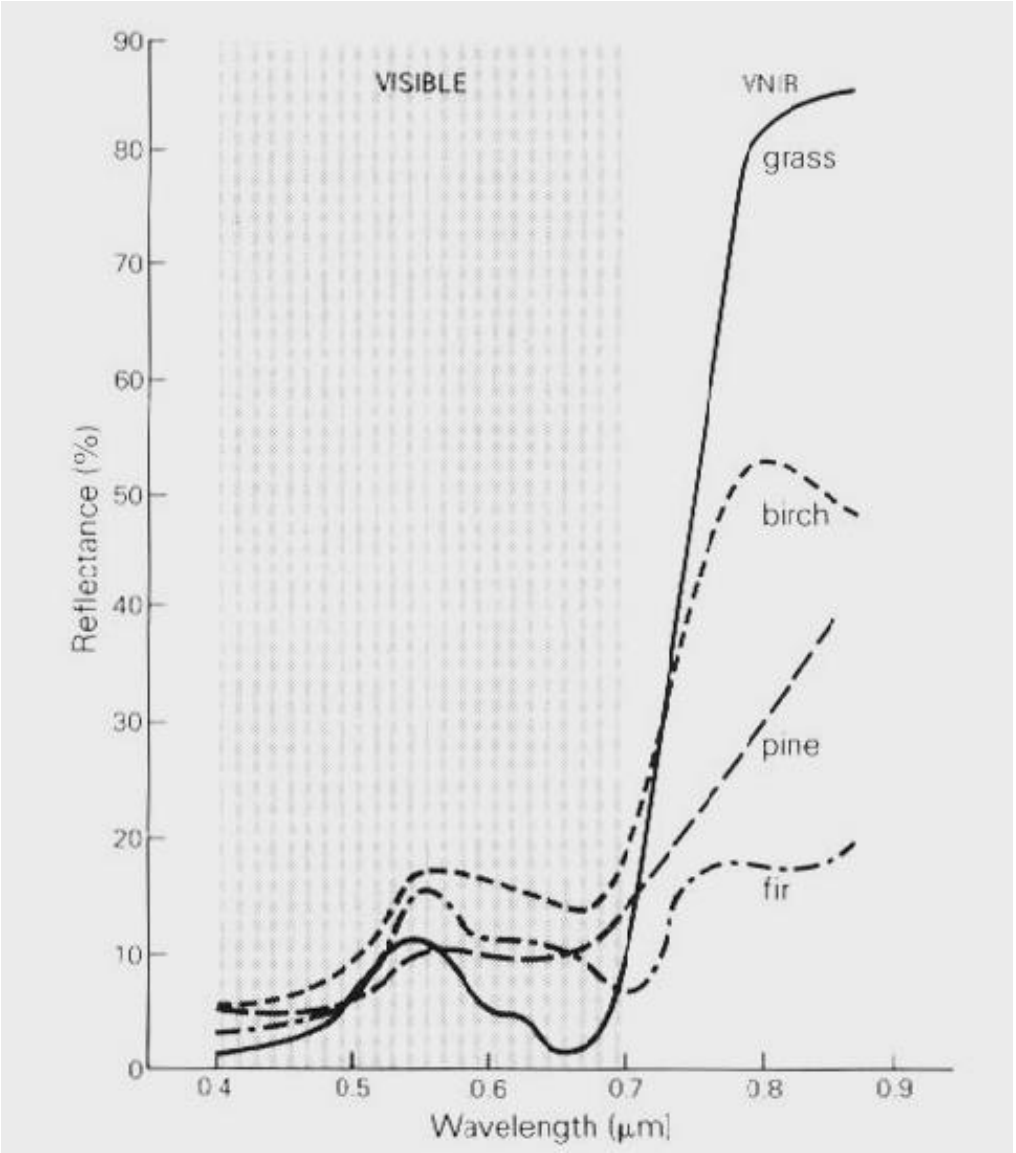
False Colour InfraRed

Vegetation Separability



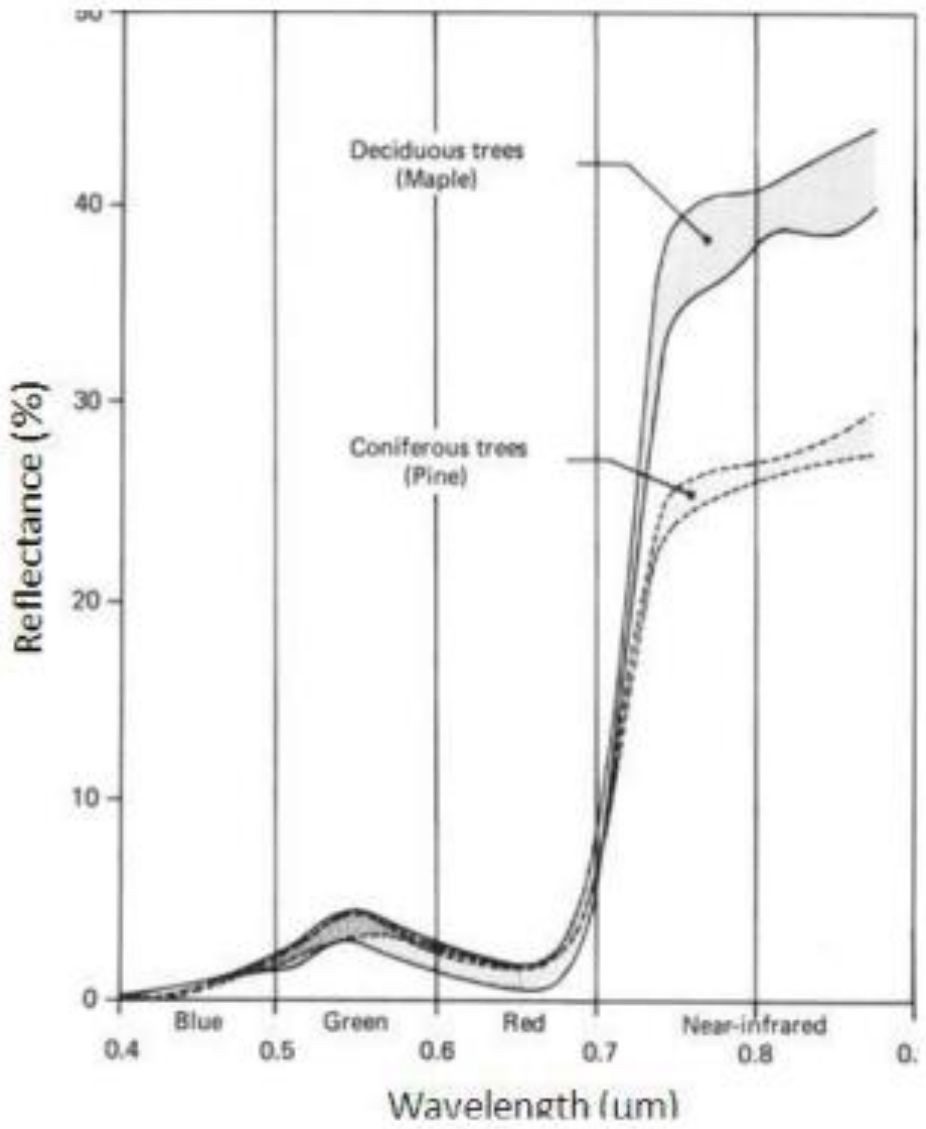
Spectral Curves- Trees

- All of the different attributes of chlorophyll content, leaf shape, area, and number, together with plant structure, contribute to spectral reflectance of plant species
- All four plants have similar responses in visible but are clearly distinguished in by their NIR reflectance



Spectral Curves- Trees

- Vegetation canopies generally display a layered structure
- Therefore, the energy transmitted by one layer is available for reflection or absorption by the layers below it
- Due to this multi-layer reflection, total infrared reflection from thicker canopies will be more compared to thin canopy cover



Spectral Curves- Crops

